



UNLOCKING THE POWER OF TMDL: ENHANCING POWER BI DEVELOPMENT

Jason
Romans

SQL Saturday Atlanta 2026 • AI
& BI

March 2026



Jason Romans

Cloud Data & Integration Developer



The DAX Shepherd



 @sql_jar

 jason-r-sql-jar

 <https://thedaxshepherd.com/>

 Nashville, TN, USA

 Began Career as a SQL Server DBA

 Transitioned to Microsoft BI Stack

 Data Engineering to Data Modeling

Standing on the Shoulders of Giants



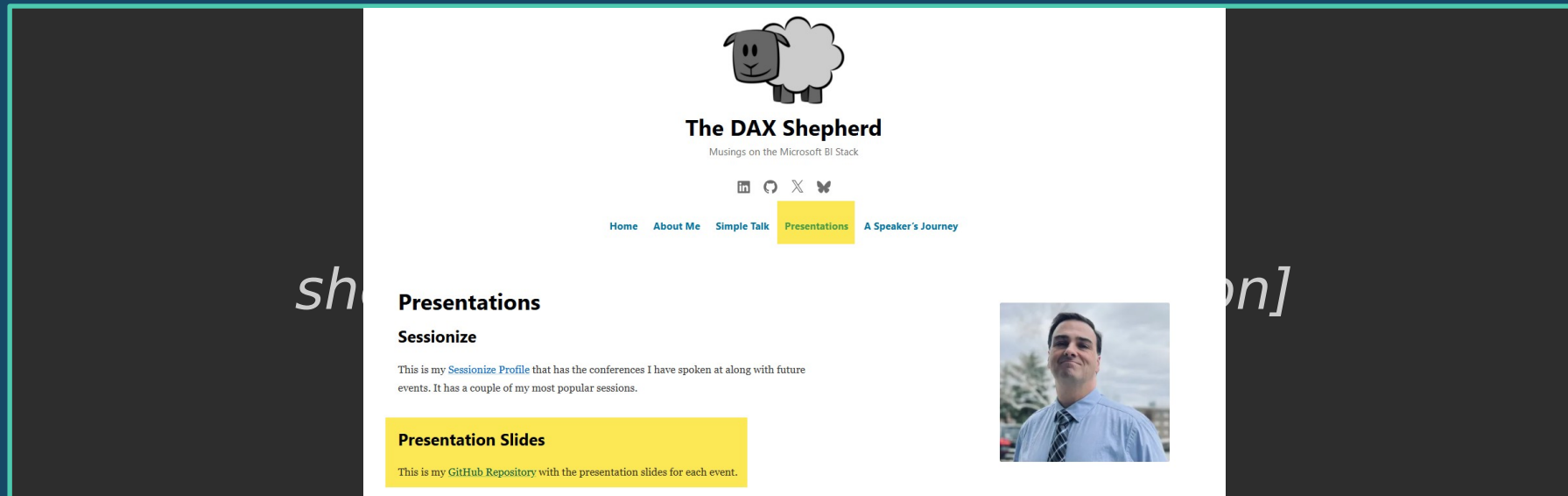
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Grateful for the people who share their knowledge, encourage us, answer questions, and help the rest of us learn faster.

DOWNLOAD THIS PRESENTATION

All slides, resources, and demo files available:

The DAX Shepherd
thedaxshepherd.com/presentations



Thank You To Our Sponsors For Making SQL Saturday Possible!



KEEP YOUR TAX DOLLARS LOCAL!

The PEACH Education Tax Credit lets you redirect your Georgia state income tax dollars directly to Innovation Academy — at no additional cost to you. This is a 100% dollar-for-dollar tax credit (not a deduction).

WHY THIS MATTERS

Instead of sending your tax dollars to the state's general fund, you can directly support:

- Mentorships & internships
- STEM and innovation programs
- Curriculum development
- Teacher professional growth
- Real-world student experiences

ELIGIBILITY LIMITS

- **Single Filer:** UP TO \$2,500
- **Filing Jointly:** Up To \$5,000
- **LLC Member or Pass Through Entity:** Up to \$25,000
- **C or S Corporation:** up to 75% of annual Georgia income tax liability

Same Taxes. Greater Impact.

STEP 1: APPLY

- Apply for the PEACH Tax Credit
- Select "FCS Innovation Academy"
- Receive approval within approximately 10 days

STEP 2: DONATE

- Follow the instructions in your approval letter
- Submit your donation (online or by mail)
- Must be completed within 60 days of approval

STEP 3: CLAIM

- File Form IT-QED-TP2 with your Georgia tax return
- Receive your full dollar-for-dollar state tax credit



gfpe.org/tax_credit/

A Very Special Thank You To Our Community Partner Innovation



**Please Support Via
Tax Credit Eligible
Donation That Goes
DIRECTLY To
Innovation Academy**

Lunch Hour Sponsor Sessions



DATALOGZ

***Sponsor Session Attendees Can Win A \$50 Amazon Gift Card Each
Two Raffle Gift Cards Per Session***



SQL Saturday Atlanta Closing Ceremony 5PM In The Auditorium Must Be Present To Win Raffle Prizes

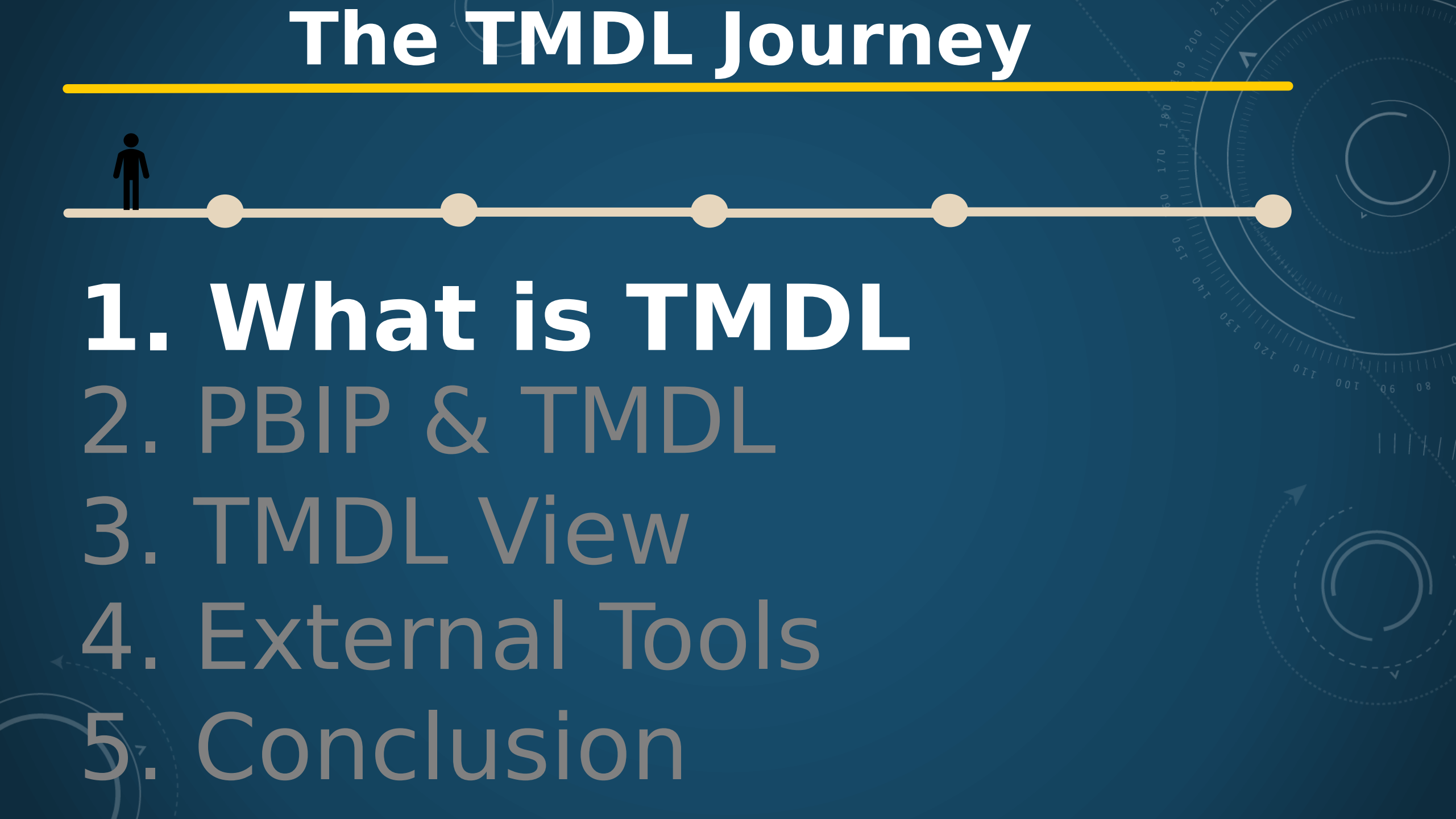


The TMDL Journey



1. What is TMDL
2. PBIP & TMDL
3. TMDL View
4. External Tools
5. Conclusion

The TMDL Journey



1. What is TMDL

2. PBIP & TMDL

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IN THE BEGINNING...

- SQL Server Analysis Services Tabular
 - Single BIM File
 - Tabular Model Scripting Language (TMSL)
 - JSON Format
- Creator of Tabular Editor - Daniel Otykier
 - TE started as a glorified JSON editor

Sample Model.Bim File

```
{ } model.bim X
Contoso10K - Copy.SemanticModel > { } model.bim > { } model > [ ] cultures > { } 0
3      "model": {
44      "relationships": [
112      }
113      ],
114      "sourceQueryCulture": "en-US",
115      "tables": [
116      {
117        "name": "Product",
118        "annotations": [
119        {
120          "name": "PBI_ResultType",
121          "value": "Table"
122        }
123        ],
124        "columns": [
125        {
126          "name": "ProductKey",
127          "annotations": [
128          {
129            "name": "SummarizationSetBy",
130            "value": "Automatic"
131          }
132          ],

```

ISSUES WITH A SINGLE LARGE BIM FILE

- Difficult to comprehend
 - Too much text is focused on delimiters {}, ()
- Single, large file - harder to collaborate with others
- Power BI and Fabric are advancing toward complete version control across all assets
 - Does it exist if it is not in source control?

POWER BI PBIX FILE

In some ways a step
backwards



POWER BI DEVELOPER MODE

- Store PBIX as a Project Folder
 - PBIP
- Enhanced Format for Semantic Model
 - TMDL
- Enhanced Format for Reports
 - PBIR

WHAT'S IN A NAME



Tabular

Model

Definition

Language

KEY POINTS

- Ease of reading – similar to YAML
- Folder structure – separate files for each table, perspective, role, and culture
 - Increases collaboration
- Minimal delimiters
- AI-ready: plain text = Copilot-friendly

WHAT IT IS NOT

THIS IS NOT IN JSON
FORMAT.



WWW.SMLCREW.COM

TMDL CREDIT

- Mathias Thierbach
 - Microsoft MVP
 - pbi-tools
- Community involvement

DUAL PURPOSE

1. Definition Language

2. Scripting Language

UNLOCKING THE POWER OF TMDL

With code:

- Create Date Table
- Turn on or off features
 - Auto Date/Time
 - Discourage Implicit Measures
- Quickly Create Objects
- Move object between Projects
- Make mass changes quickly

Example of TMDL

```
1 createOrReplace
2
3   table Date
4     lineageTag: 093cc89d-ab2b-423e-9810-be9d879345e1
5
6     column Date
7       dataType: dateTime
8       formatString: Long Date
9       lineageTag: 0bbfcfc4-20b3-48c4-b1d8-b3580bde1eee
10      summarizeBy: none
11      sourceColumn: Date
12
13      annotation SummarizationSetBy = Automatic
14
15      annotation UnderlyingDateTimeDataType = Date
16
17    column Year
18      dataType: int64
19      formatString: 0
20      lineageTag: 7de18a58-8189-4a64-a8b9-9b278f40a9d2
21      summarizeBy: sum
22      sourceColumn: Year
23
```

Quick TMDL Primer

- Whitespace for indenting
- Hierarchy
 - Table
 - Column
 - Properties
 - Measure
 - Properties



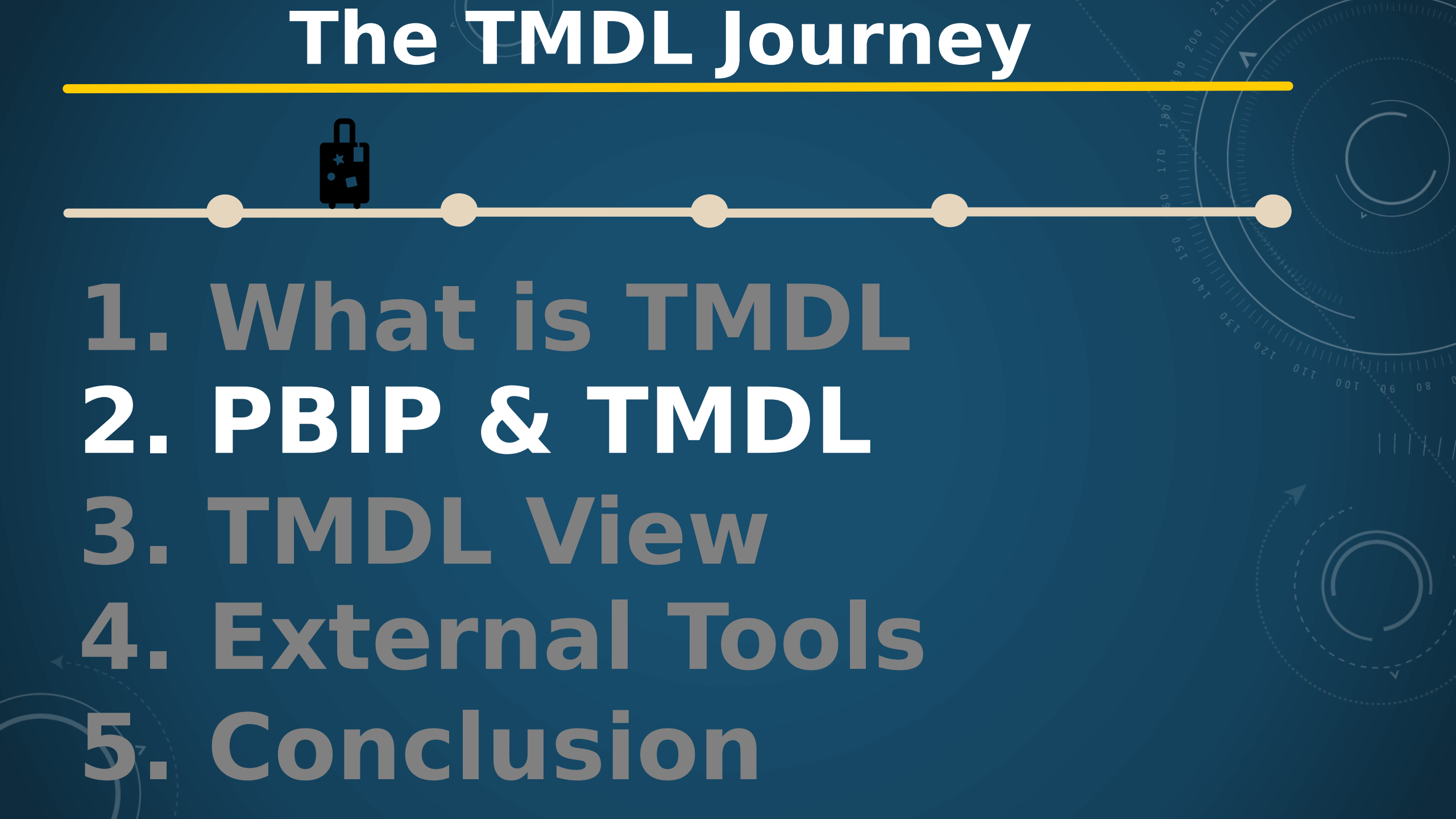
More TMDL

- Triple slash (///) syntax
 - Description – good for documentation
 - For example /// Table Description

Best Way to Learn Syntax

- Create Object in GUI
 - Script out object
- View snippets

The TMDL Journey



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TMDL IN POWER BI DESKTOP

Related but independent

- Power BI Project (.pbip)
 - Store semantic model in TMDL format
- TMDL View
 - Can be used with PBIX files

BREAKING NEWS - READ ALL ABOUT IT

- September 2025 Power BI Released
 - TMDL View is GA
 - [Power BI September 2025 Feature Summary](#)

PREVIEW OPTIONS - POWER BI PROJECT

- ☒ Power BI Project (.pbip) save option [Learn more](#)
 - ☒ Store semantic model using TMDL format [Learn more](#)
 - ☐ Store reports using enhanced metadata format (PBIR) [Learn more](#)
 - ☐ Store PBIX reports using enhanced metadata format (PBIR) [Learn more](#)

LEARN MORE

- ☒ Power BI Project (.pbip) save option [Learn more](#) 
- ☒ Store semantic model using TMDL format [Learn more](#)
- ☐ Store reports using enhanced metadata format (PBIR) [Learn more](#)
- ☐ Store PBIX reports using enhanced metadata format (PBIR) [Learn more](#)

TMDL IN POWER BI DESKTOP

1. Power BI Project (.pbip)

- Store semantic model in TMDL format

FOLDER RENAMING IS NOT VERSION CONTROL

Name	Status	Date modified	Type	Size
 BeforeBossFight_Original_Power_BI		3/20/2025 9:23 AM	File folder	
 Broken_Original_Power_BI		3/20/2025 9:23 AM	File folder	
 Lost_Original_Power_BI		3/20/2025 9:24 AM	File folder	
 Original_Power_BI		3/20/2025 9:22 AM	File folder	
 Restore_of_Restore_Original_Power_BI		3/20/2025 9:24 AM	File folder	
 zOriginal_Power_BI		3/20/2025 9:22 AM	File folder	



VISUAL STUDIO CODE DEMO

POWER BI – TMDL ISSUE

Issues were found

There's a problem with the definition content in your Power BI Project.

TMDL Format Error:

Parsing error type - UnsupportedObjectType

Detailed error - Unsupported object type - 1model is not a supported property in the current context!

Document - './model'

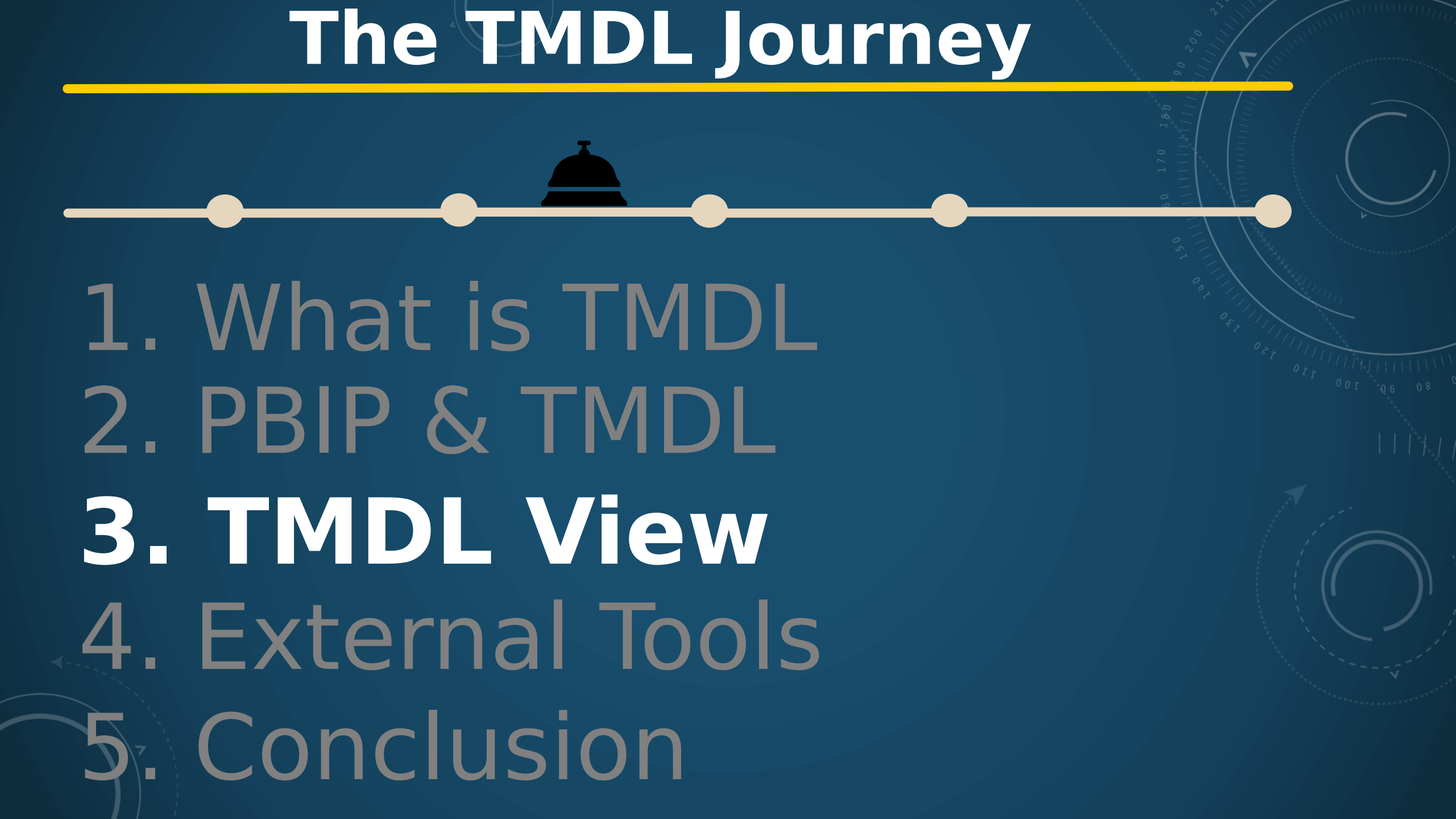
Line Number - 2

Line - '1model Model'

[Copy details to clipboard](#)

Close

The TMDL Journey



1. What is TMDL

2. PBIP & TMDL

3. TMDL View

4. External Tools

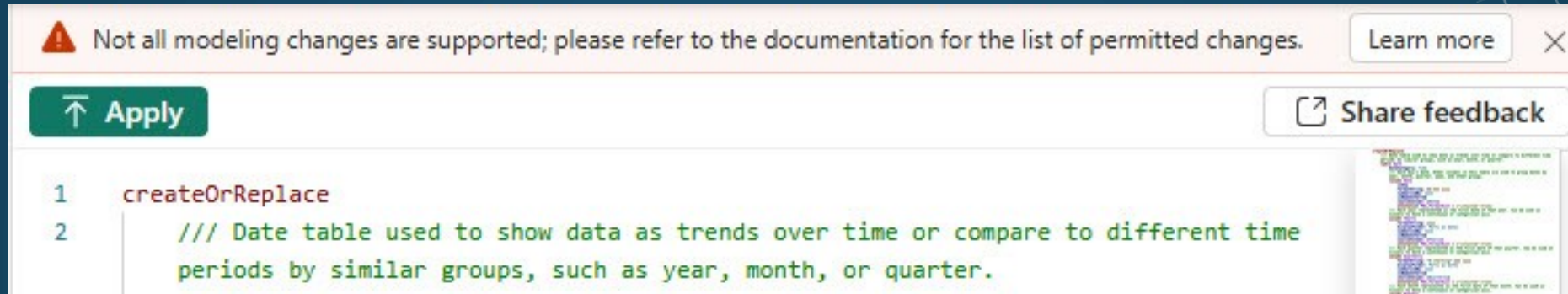
5. Conclusion

TMDL IN POWER BI DESKTOP

2.TMDL View

- Can be used with PBIX files

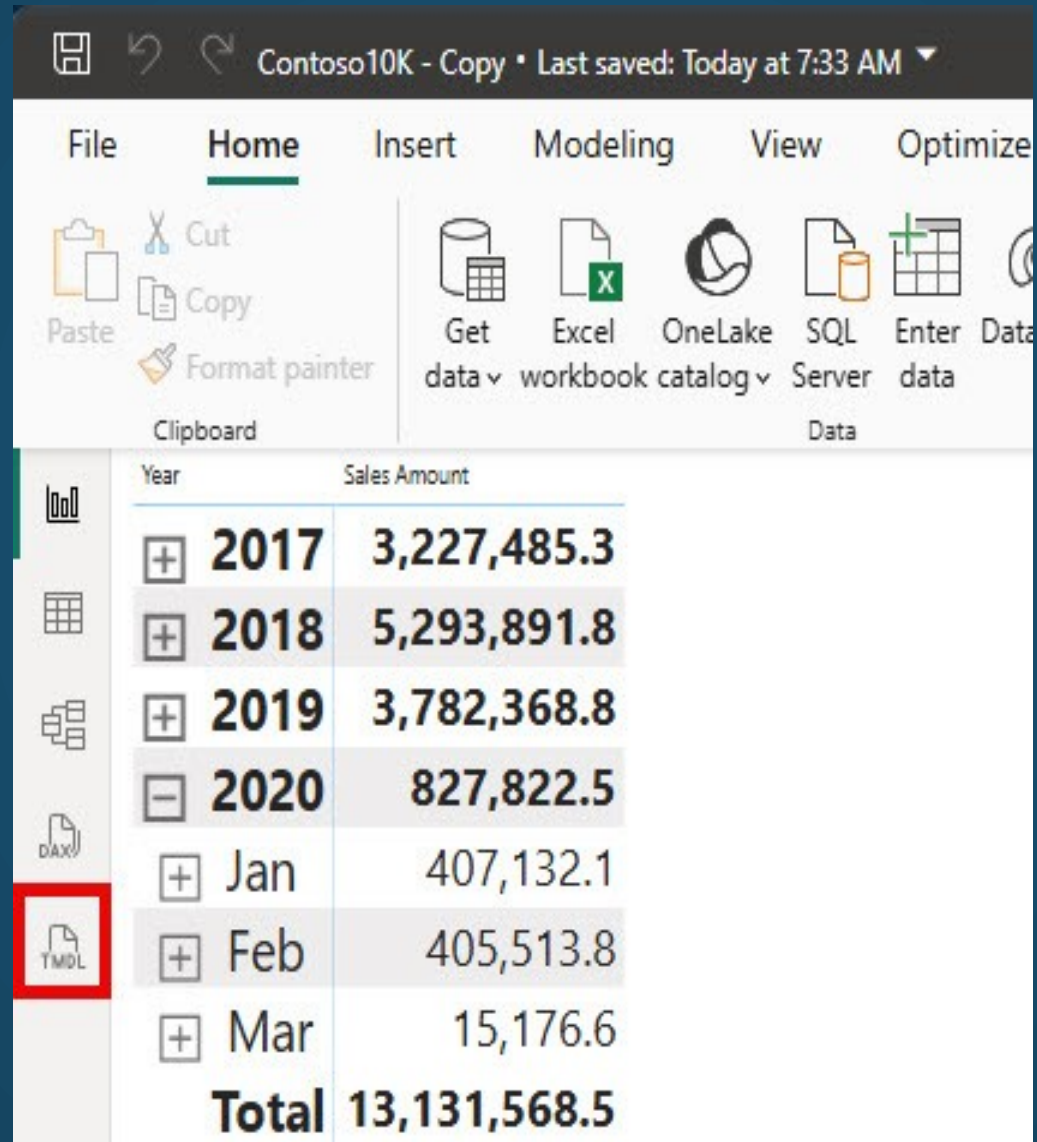
WARNING



Learn More:

<https://learn.microsoft.com/en-us/power-bi/developer/projects/projects-overview#model-authoring>

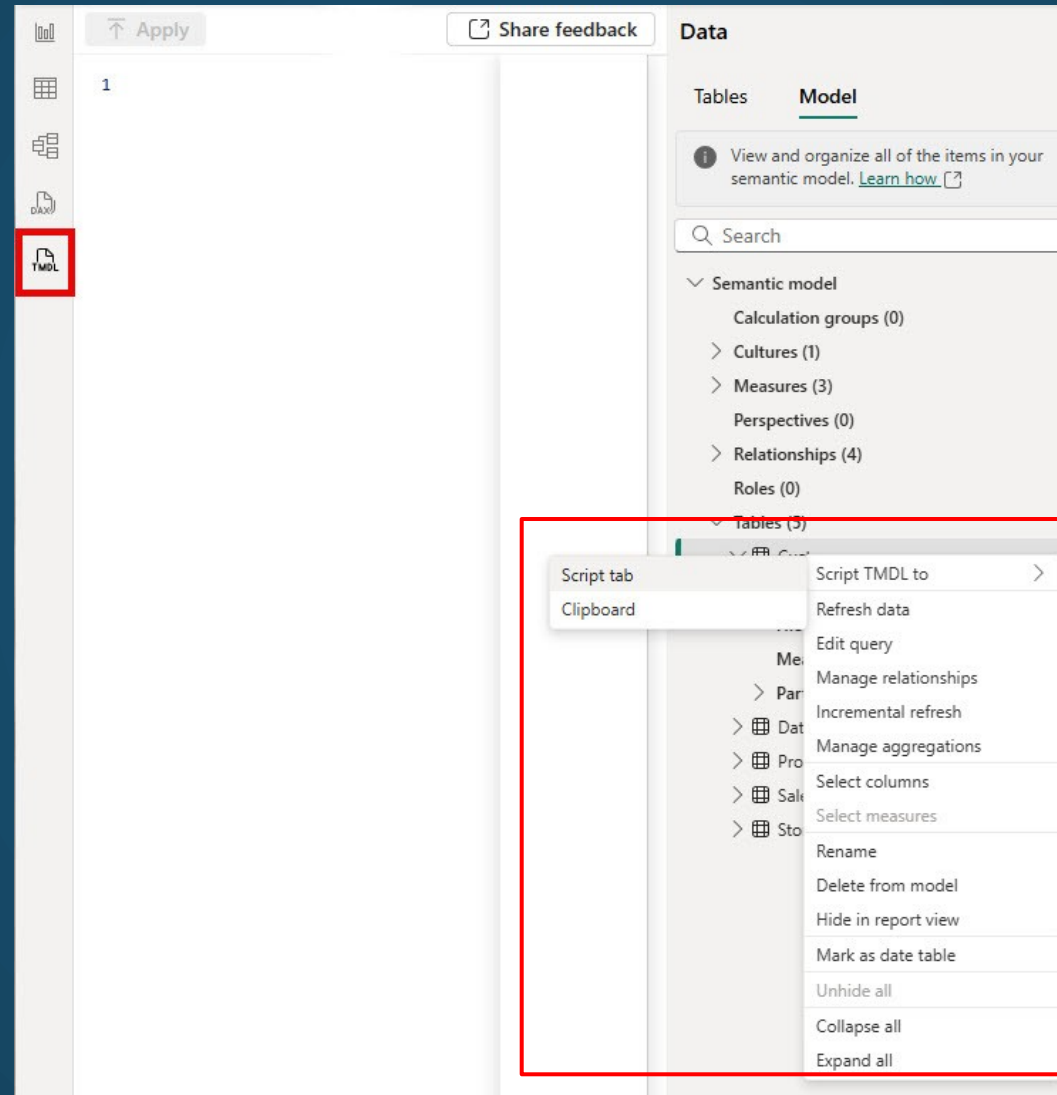
TMDL VIEW BUTTON



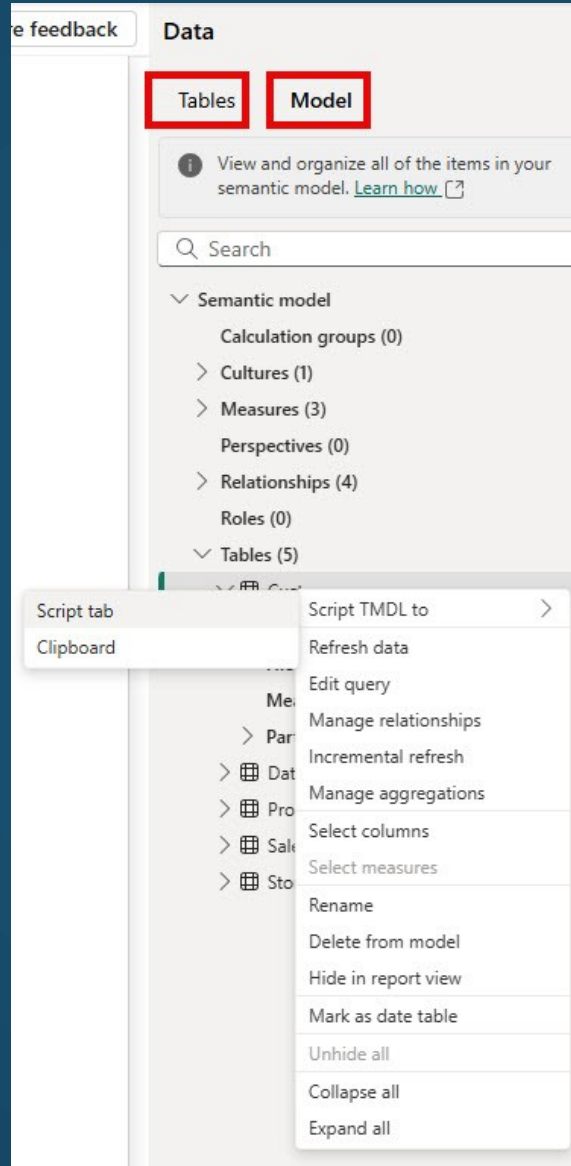
The screenshot shows the Microsoft Power BI Desktop interface. The ribbon at the top includes tabs for File, Home, Insert, Modeling, View, and Optimize. The Home tab is active, showing groups for Clipboard (Paste, Cut, Copy, Format painter) and Data (Get data, Excel workbook, OneLake catalog, SQL Server, Enter data). On the left sidebar, the TMDL view button is highlighted with a red box. The main area displays a table with sales data.

	Year	Sales Amount
+	2017	3,227,485.3
+	2018	5,293,891.8
+	2019	3,782,368.8
-	2020	827,822.5
+	Jan	407,132.1
+	Feb	405,513.8
+	Mar	15,176.6
	Total	13,131,568.5

SCRIPT TMDL

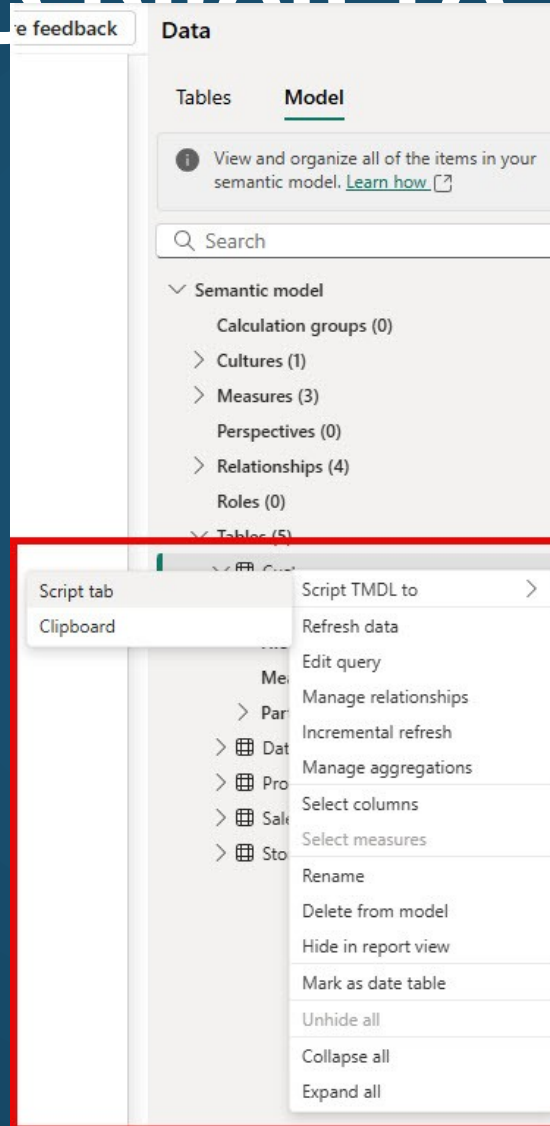


CAN BE IN TABLE OR MODEL SELECTION

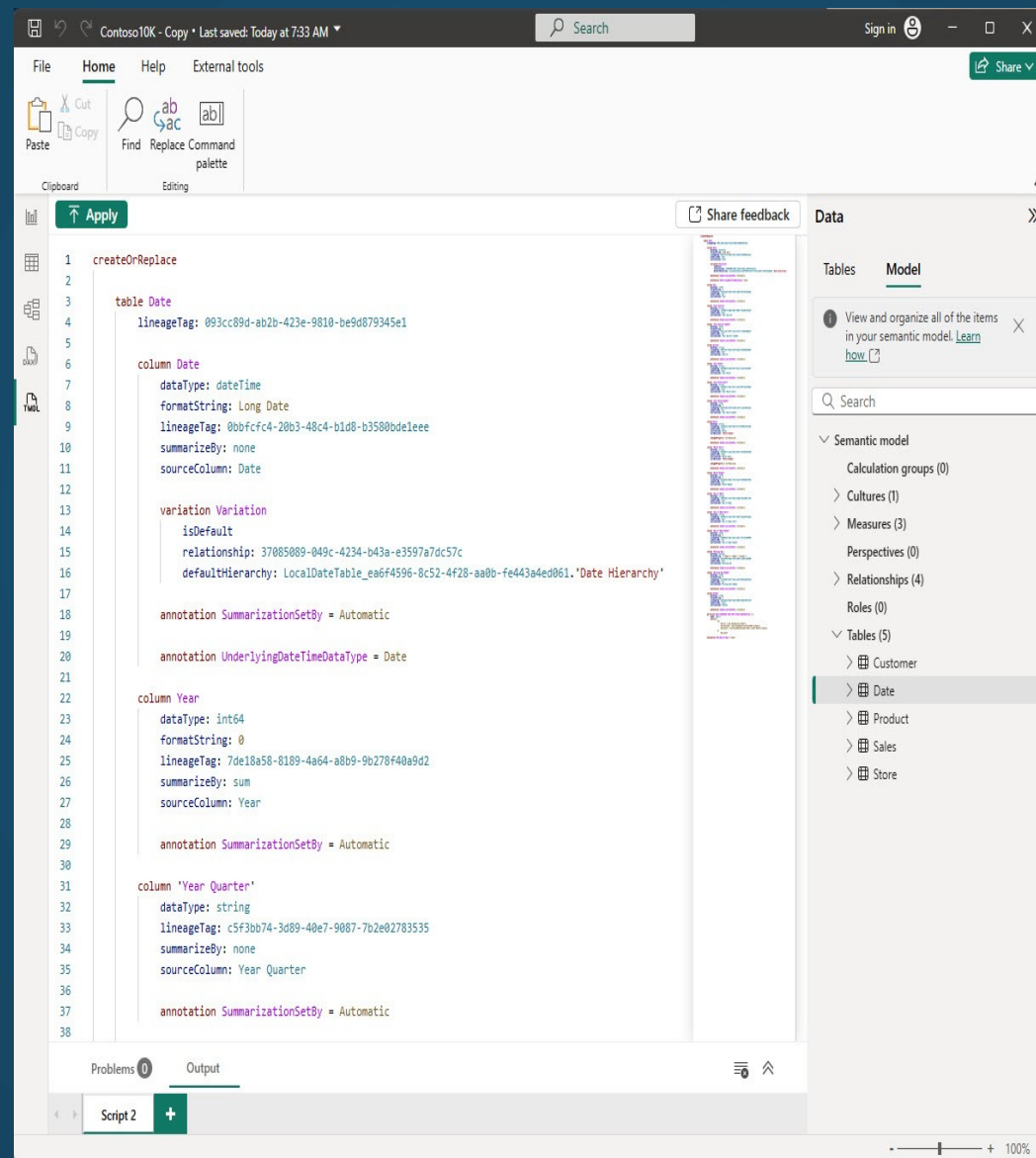


CAN SCRIPT TO SCRIPT TAB OR CLIPBOARD

ALSO, DRAG AND DROP TO TAB



TMDL VIEW – SCRIPT TAB



TMDL VIEW – CODE INTERFACE

↑ Apply

Preview

🔗 Share feedback

1 createOrReplace

2

3 table Customer

4 lineageTag: 8b57b73a-c7ab-4e11-bd1d-3b4134f84e05

5

6 column CustomerKey

7 dataType: int64

8 formatString: 0

9 lineageTag: 476a899b-b9ab-4699-bba8-df9642e39bfd

10 summarizeBy: none

11 sourceColumn: CustomerKey

12

13 annotation SummarizationSetBy = Automatic

14

15 column Gender

16 dataType: string

17 lineageTag: a373a8a9-7a80-4f5a-852b-0374aef50f7e

18 summarizeBy: none

19 sourceColumn: Gender

20

21 annotation SummarizationSetBy = Automatic

22

23 column Name

24 dataType: string

25 lineageTag: 3cfe8fb1-58aa-4087-9781-a6a68a5ed24e

26 summarizeBy: none

27 sourceColumn: Name

28

29 annotation SummarizationSetBy = Automatic

30

31 column Address

Problems 0

Output

⌵ ⌴

Script_CustomerTable

+

CODE INTERFACE - PREVIEW

↑ Apply Preview

Share feedback

↑ ↓ ⌕ ... ✕

1 createOrReplace

2

3 table Customer

4 lineageTag: 8b57b73a-c7ab-4e11-bd1d-3b4134f84e05

5

6 column CustomerKey

7 dataType: string

8 formatString: 0

9 lineageTag: 476a899b-b9ab-4699-bba8-df9642e39bfd

10 summarizeBy: none

11 sourceColumn: CustomerKey

12

13 annotation SummarizationSetBy = Automatic

14

15

16 column Gender

17 dataType: string

18 lineageTag: a373a8a9-7a80-4f5a-852b-0374aef50f7e

19 summarizeBy: none

20 sourceColumn: Gender

21

22 annotation SummarizationSetBy = Automatic

23

24 column Name

25 dataType: string

26 lineageTag: 3cfe8fb1-58aa-4087-9781-a6a68a5ed24e

27 summarizeBy: none

28 sourceColumn: Name

29

30 annotation SummarizationSetBy = Automatic

31

32 column Address

33 dataType: string

34 lineageTag: 61be27ea-9431-48fb-956d-7e3c265d0c5f

35

1 model Model

419 table Date

575 partition

577 source =

582 in

584

585 annotation PBI_ResultType =

Table

586

587 table Customer

588 lineageTag:

8b57b73a-c7ab-4e11-bd1d-3b4134

f84e05

589

590 column CustomerKey

591 dataType: int64

592 formatString: 0

593 lineageTag:

476a899b-b9ab-4699-bba8-df

9642e39bfd

594 summarizeBy: none

595 sourceColumn: CustomerKey

596

597 annotation

SummarizationSetBy =

Automatic

598

599 column Gender

600 dataType: string

601 lineageTag:

a373a8a9-7a80-4f5a-852b-03

74aef50f7e

1 model Model

419 table Date

575 partition

577 source =

582 in

584

585 annotation PBI_ResultType = Table

586

587 table Customer

588 lineageTag:

8b57b73a-c7ab-4e11-bd1d-3b4134f84

e05

589

590 column CustomerKey

591+ dataType: string

592 formatString: 0

593 lineageTag:

476a899b-b9ab-4699-bba8-df964

2e39bfd

594 summarizeBy: none

595 sourceColumn: CustomerKey

596

597 annotation

SummarizationSetBy =

Automatic

598

599 column Gender

600 dataType: string

601 lineageTag:

a373a8a9-7a80-4f5a-852b-0374a

ef50f7e

Problems 0 Output

Script_CustomerTable +

TMDL VIEW – RENAME TABS

```
28      annotation PBI_FormatHint = {"isCustom":true}
29      /// Each month represented as the first date of that
      month. Can be used on visuals to have a continuous or
      categorical axis.
30      column Monthly
31          formatString: mmm yyyy
32          displayFolder: Parts as dates
33          summarizeBy: none
```

Problems 0

Output

Date Table

Calculation Groups



TMDL SCRIPTS - SAVED IN FILE (PBIP/PBIX)

Name	Date modified	Type	Size
 .pbi	3/17/2025 8:41 AM	File folder	
 DAXQueries	3/17/2025 8:41 AM	File folder	
 TMDLScripts	3/17/2025 8:41 AM	File folder	
 .platform	3/17/2025 8:41 AM	PLATFORM File	1 KB
 definition.pbism	3/17/2025 8:41 AM	PBISM File	1 KB
 diagramLayout.json	3/17/2025 8:41 AM	JSON File	2 KB
 model.bim	3/17/2025 8:41 AM	BIM File	79 KB

ENABLES FUNCTIONALITIES WITHOUT EXTERNAL TOOL

For example:

- Perspectives
- Translations

Share TMDL Scripts

- Centralized collection
 - SharePoint site or Teams
- Sites with TMDL samples
 - TMDL Gallery
 - FabSnippets
- Article in resources regarding security precautions



TMDL VIEW DEMO



The TMDL Journey



1. What is TMDL
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- 4. External Tools**
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EXTERNAL TOOLS

- Visual Studio Code
 - TMDL Extension
 - Power BI Studio
- pbi-tools
- Tabular Editor

TABULAR EDITOR 2.X – SAVE TO FOLDER – TMDL

Tabular Editor Preferences

Features General DAX **Serialization**

Serialization mode: TMDL ☒ Save .bim copy

General Serialization Settings

☒ Ignore inferred objects	☒ Ignore inferred properties
☒ Ignore timestamps	☒ Split multiline string
☐ Ignore lineage tags	☐ Ignore privacy settings
☐ Ignore incr. refresh partitions	☐ Include sensitive

TMDL Serialization Options

☐ Include metadata order

Indentation: Tabs

Indentation width: 4

Encoding: UTF8

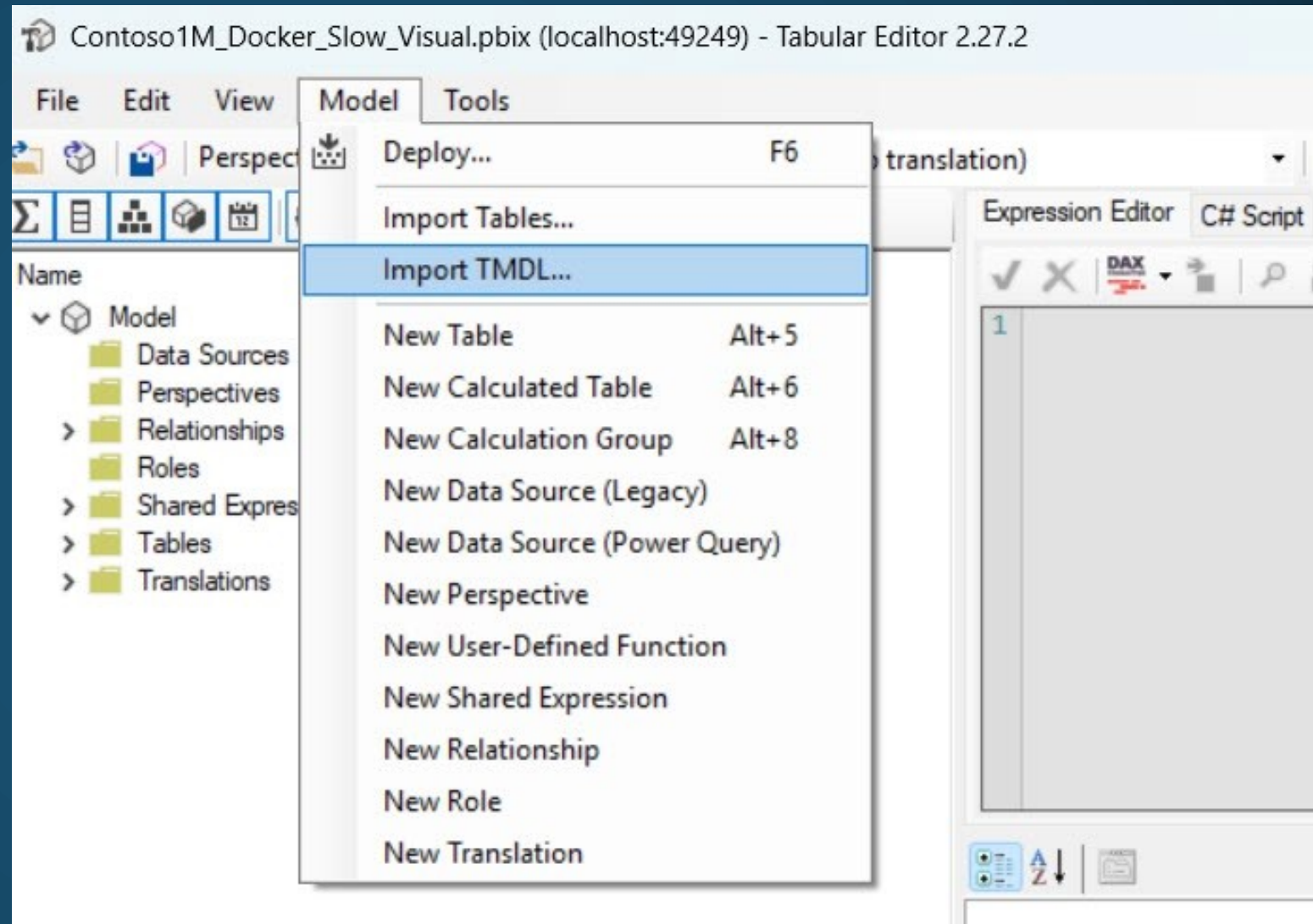
Newline style: Default

Casing style: camelCase

Expression trim style: Trim trailing whitespaces

OK Cancel

TABULAR EDITOR 2.X – IMPORT TMDL



TABULAR EDITOR

3

Preferences

- ▼ Data Browsing
 - Pivot Grid
 - DAX Query
 - Table Preview
- ▼ Text Editors
 - General
 - > DAX Editor
 - > SQL Editor
 - > M Editor
 - > C# Editor
- Schema Compare
- ▼ File Formats
 - General
 - Save-to-folder**
 - VertiPaq Analyzer
- ▼ Model Deployment
 - General
 - Data Refresh
- ▼ DAX Optimizer
 - General

Save-to-folder

Serialization mode

TMDL settings

☐ Include metadata order

Indentation

Indentation width

Encoding

Newline style

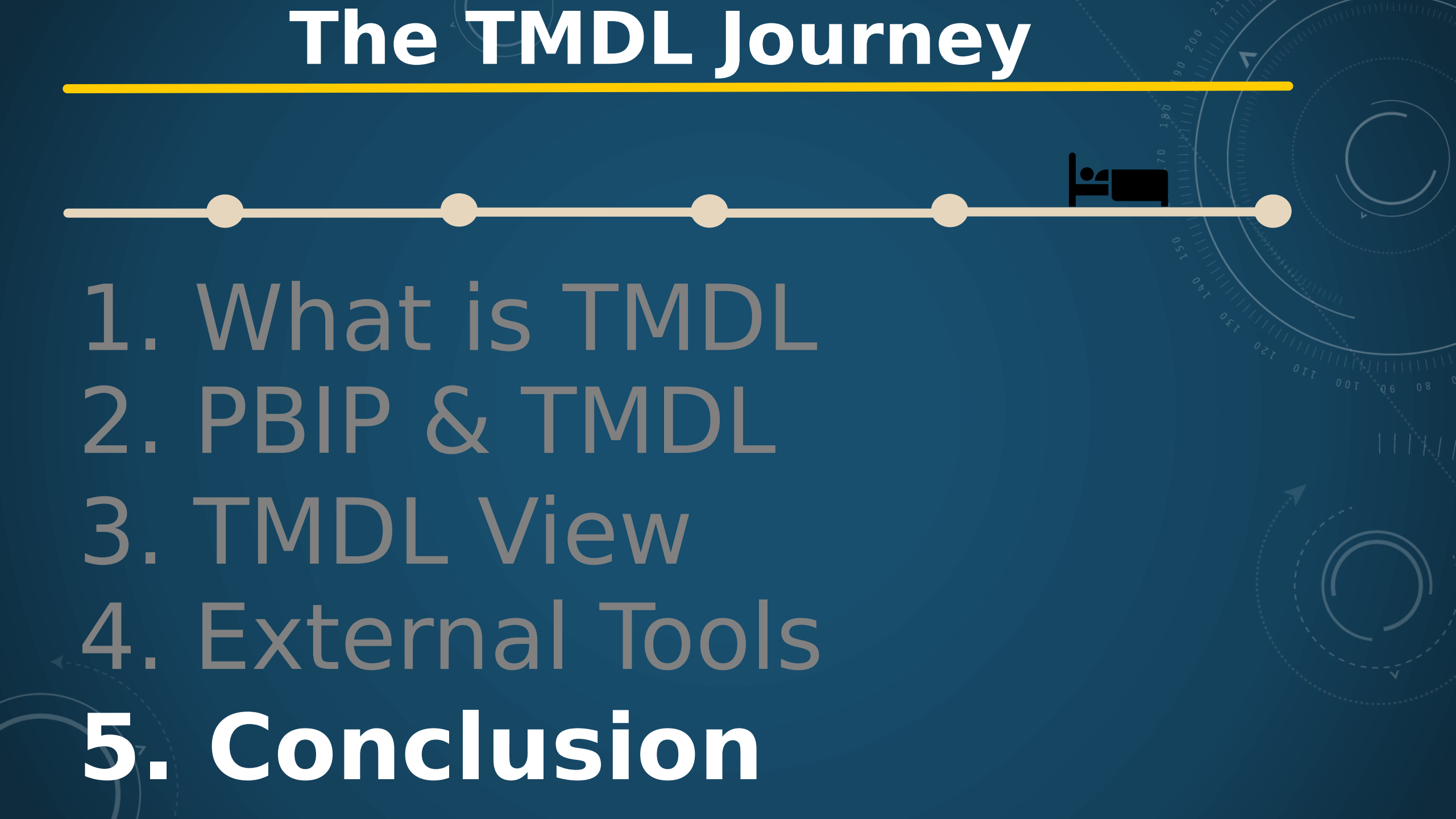
Casing style

Expression trim style

TABULAR EDITOR 2 - SAVE TO FOLDER

Name	Date modified	Type
 cultures	3/14/2025 10:48 AM	File folder
 tables	3/14/2025 10:48 AM	File folder
 database.tmdl	3/14/2025 10:49 AM	TMDL File
 model.tmdl	3/14/2025 10:49 AM	TMDL File
 relationships.tmdl	3/14/2025 10:49 AM	TMDL File

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FUTURE EXPANSION

- Expanding support for changes that are not currently supported
- TMDL View available on the Web
- TMDL View Copilot

ENDLESS POSSIBILITIES

- Assist with Development
 - Reuse Modules
- Build Models with AI
- Best Practices

RESOURCES

- Microsoft TMDL Overview
 - <https://learn.microsoft.com/en-us/analysis-services/tmdl/tmdl-overview>
- Developer Mode
 - <https://learn.microsoft.com/en-us/power-bi/developer/projects/projects-dataset#tmdl-format>
- TMDL View
 - <https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-tmdl-view>

RESOURCES

- Power BI TMDL View: Benefits, Security Risks and Best Practices by John Kerski
 - <https://www.red-gate.com/simple-talk/business-intelligence/power-bi-tmdl-benefits-security-risks-best-practices>
- FabSnippets
 - <https://fabsnippets.com>
- DataZoe – DAX Date table in TMDL
 - <https://www.datazoe.blog/post/dax-date-table-in-tmdl>

RESOURCES

- Deep dive into TMDL view for Power BI Desktop
 - <https://powerbi.microsoft.com/en-us/blog/deep-dive-into-tmdl-view-for-power-bi-desktop-pre-view/>
- Microsoft Lineage Tags
 - <https://learn.microsoft.com/en-us/analysis-services/tom/lineage-tags-for-power-bi-semantic-models>
- #SML (Saturday Morning Learning)
 - www.smlcrew.com

Hope you enjoyed
your stay

HOTEL TMDL

Check-Out Receipt

Guest: SQL Saturday Attendee
Room: The PBIP Suite

TMDL Orientation.....FREE
Source Control Access...FREE
TMDL View (GA).....FREE
Tabular Editor Tips.....FREE
Late Checkout..NOT SUPPORTED

TOTAL:.....FREE

*Thank you for staying at Hotel TMDL
— a YMCA property —*

Session Evaluations



SQL Saturday Atlanta, AI & BI -
Session Evaluation



Thank you Jason Romans



thedaxshepherd@gmail.com



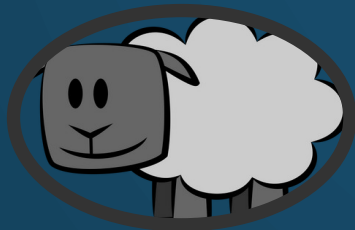
jason-r-sql-jar



<https://thedaxshepherd.com/>



@sql_jar



The Dax Shepherd



Contact
Information